ALL-INK

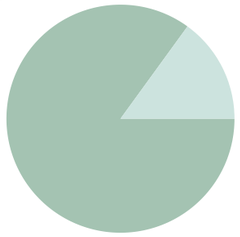


The new ink cartridge by ALL-INK offers a refillable alternative to the typical classic single-use ink cartridge. ALL-INK meticulously designed every aspect, from interlocking system to service system, with a focus on sustainability by integrating circular principles

Group N:
Emmanuel Irechukwu
Arancha Ramirez
Chuwei Xiong

PROBLEM & CAUSATION

Identifying problems found in the ink cartridge industry and recycling system and exploring how our product and service system could solve them.



85%

ink cartridges are NOT recyled [1]

Consumer Pain Points

Buying a new cartridge every time adds an extra cost

It can feel unnecessary and expensive having to purchase a new consumable every time, wasting time having to wait or travel to acquire it.

Hassle of buying & recycling ink cartridges regularly

Remembering to dispose of your cartridge every time is inconvenient. It is a hassle as it requires trips to the post office. Which in turn, induces transport emissions.

Confusion over different recycling systems

Most consumers struggle with adequately sorting and sending their ink cartridges to the appropriate recycling schemes.

System Solutions

Take-back service

ALL-INK streamlines cartridge returns. Our user friendly program eliminates costly mail returns as the ink cartridges are collected door to door.

Collaboration

The brand of the ink cartridge does not matter as ALLL-INK use any type of ink cartridge, as it is one whole system the issue of brand ideals not aligning does not matter..

Distribution

Ink cartridges will be refilled in the UK and will be distributed efficiently with a combination of delivery-collection service involving ALL-INK mail box

Communication

Updates and information to user through our app will raise awareness about the impact ink cartridges have and how to reduce this. Additionally, it will notify users on their progress and contribution to the system's circularity.

Recycling Systems

All ink cartridges brands will be accepted into the refilling system

"350 million consumables end up in landfills each year" [1]

PROBLEM AREAS

Current cartridge return methods can be inconvenient and costly for users who have to send them back through mail, often without subsidies.

Ink cartridges are difficult to take apart and messy to refill due to strong glue adhesion

Brands have different goals, which makes it difficult to come up with whole system solutions as views are not aligned

Most brand's recycling schemes take place outside the UK , so ink cartridges have to be shipped back and forth

Lack of communication about ink cartridges recycling schemes between manufacturers and consumers

Ink cartridges are difficult to recycled due to mixed plastic materials used in the casing

Foam in the ink cartridge can't be recycled and will most likely end up in landfills

Ink cartridges are sold wrapped in excessive, wasteful packaging protective layers, many of which are made of un-recyclable plastic

Brands only accept their own consumables into their recycling schemes

There is no incentive for users to recycle their ink cartridges, which may be why only 15% of ink cartridges get recycled

Product Solutions

Dissassembly

Ink cartridge lids will be altered to allow for easy access when needing to refill ink cartridges, making the refilling process more effiecient.

Disposal

Instead of disposing of the ink cartridges the ink cartridges will be reused meaning the issue of disposal because of mixed plastic materials is not an issue.

Material Design

ALL-INK plan to utilise used foam from ink cartridges that can't be recovered anymore as a protective layer for the plastic boxes outside of users houses where they can place their empty ink cartridges.

Packaging Design

Ink cartridges will be deposited into padded plastic boxes that will be placed outside consumers houses, meaning nothing except the ink cartridges are supplied to users.

User Motivation

There will be a point system that users can collect as they return ink cartridges if they collect enough point their subscription price for the next month may be reduced.

FUTURE SCENARIO & TIME HORIZON

Defining our solution and analysing the specific time frame where our product will be implemented and have the most impact.

Our Solution

ALL-INK produces a novel, 100% recycled plastic cartridge, designed to reduce the cartridge pollution and improve sustainability in the next **5-10 years**, where sustainability and circular economy is at the core of people's mindset**.** Its removable cap, connected by an interlocking system to the body, allows for the cartridge to be refilled in our facilities.

Our system is subscription based and is completely maintained through an online platform. When the user subscribes to our product, they supply an empty cartridge to be modified and recycled or request an already modified cartridge to be provided. A company's delivery box is installed outside the user's home to allow protected and package-free deliveries. The altered cartridge is then transported and delivered directly to the user via online delivery services. The user requests a new cartridge by scheduling an ink delivery via the app, once he's notified of low ink levels. The users deposit the empty cartridge in the provided box to be retrieved by the delivery service. The used cartridges are returned to the company, where they are tested, sanitized and refilled so that they can be reused.

Time Horizon

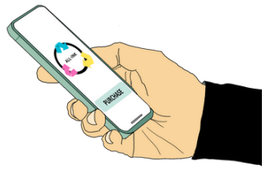
Sourcing Material for lid

350 million consumables end up in landfills each year whose plastic components take up to 1000 years to decompose [1], so reducing the amount of cartridges in produced is imperative. Our lid is made from recycled cartridge plastic sourced from our recycling scheme and aluminum is used for the metal interlocking mechanism.



Online Platform

Our system will rely heavily on an online platform. The app is used as a medium for the users to subscribe to our product, keep track of their ink levels and request and schedule deliveries to their doorstep. Our goal is to provide a tailored and attentive experience to the user to earn their loyalty towards our brand.



Expansion



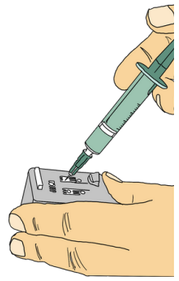
Drone Delivery Service

In order to reduce transportation impact and ensure prompt and efficient arrival, our products will be delivered with the help of drones. This will also increase accessibility to system and simplify repacking and resending the consumable when it has reached the end of its use cycle

Bio-material Ink

Alternative non-polluting or non-hazardous ink will replace the current oil-based ink. This will reduce the need for foam reservoirs while maintaining the ink quality.

2023



It is essential for the cartridge to be designed for disassembly to easily access the ink chamber during the refilling process. The interlocking mechanism of the lid to the body, ensures clean and secure access, avoiding ink spillages.

Design for Refilling



Our brand will send their products packaging free so that consumers don't have to dispose of the extra material, which avoids unboxing our product becoming a pain point. Users receive the cartridges in the previously installed mail boxed who's inside is lined with 100% cartridge recycled foam to provide protection for the cartridges delicate electrical components

Packaging-free Product

2033

OUR SYSTEMS DIAGRAM

ALL-INK system described in detail.

Initially, user subscribes to the service system through the website. Their order is placed to receive new ink cartridges or they provide empty ones and the company's mail box is installed in their home by ALL-INK. The user then downloads the ALL-INK app to manage their ink cartridge consumption.

INFRASTRUCTURE
DATA
VALUE
RESOURCE STATES
PHASES: manufacturing | delivery | recoverer | consumption

Material Sourcing

cartridge materials
- Main material: 100% recycled
- Brand recovers PS pellets

water+pigment+ chemical additives
- Main material: 100% virgin
- Brand buys ink materials

aluminium+nickel
- ALL-INK buys aluminium +nickel

STAKEHOLDER: manufacturing worker

suppliable

manufacturer

cartridge case + lid production
Method: injection molding
ink formulation
Method: grinding +mixing

assembly

Ink foam is stored inside case & case + lid are glued together

delivery

ALL-INK purchases recycled cartridges form brand in case user can't provide empty ones

STAKEHOLDER: refilling facility worker

cap alterations
- Injection molding + die cast to adapt lid
  Case + lid attach together through interlocking mechanism, only separable by manufacturer
- Information on how subscription works + instructions on app

distribution ready

storage

refill service

cap alterations
- Injection molding + die cast to adapt lid
  Case + lid attach together through interlocking mechanism, only separable by manufacturer

recoverable

recycling

Damaged ink cartridges are returned to original manufacturer where they are sorted & polystyrene ether is shredded into pellets to reuse for cartridge cases

Aim: We hope for a 1 year cycle, with every cartridge being used up to 12 times before remanufacturing . An average user will need 1 ink cartridge every month.

quality test
- Durability & state of print head & nozzles are tested
- Unique cartridge code helps sort & identify cartridges

reusable

clean
- Head disassembled by manufacturer
  Cleaning solutions used to clear out dried or old ink from the cartridge's internal components
- info on how many times the cartridge has been refilled

STAKEHOLDER: driver

distribution service
- Subscription info + arrival details selected on app
- Distribution service

delivery
- Customer already paid online

Packaging: cartridges are transported in cardboard boxes that have compartments for each individual ink cartridge. Cartridges are delivered bare and deposited in protective ALL-INK box outside customer's home.

STAKEHOLDER: driver

transportation
- Distribution service is the same as the delivery

STAKEHOLDER: user

use: print
- Information about how to place ink cartridge

low ink level warning

ink is consumed
- ALL-INK app notifies user
- ALL-INK app keeps track of user's printing trends

schedule ink delivery
- Arrival details selected on app

obsolete

empty ink cartridge
- ALL_INK app provides information on recycling scheme

deposit in box
- ALL_INK app reminds users to return their empty cartridge

collection service
- Delivery service leaves new ink cartridge & takes empty one back to manufacturer
- Each cartridge has unique identification code

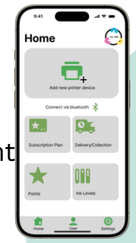
Emmanuel Irechukwu

INFRASTRUCTURE SYSTEM

All the infrastructure that ALL-INK use is detailed below, showing the interactions between the different infrastructure components.

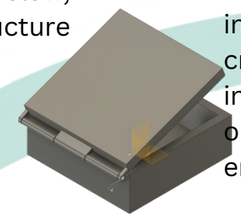
ALL-INK App

The ALL-INK app simplifies ink cartridge subscriptions from choosing plans to real-time delivery tracking. With a user-friendly interface, a loyalty point system, and strong privacy measures, it ensures a reliable and eco-conscious printing solution.



ALL-INK Box

Subscribing to ALL-INK includes on-site installation of a secure collection box by our crew. They assist with verification, introduce users to recycling benefits and our point system, fostering confidence and environmental awareness.



Delivery/Collection Box

Cardboard boxes, filled with recycled foam from prior cartridges, are employed for the delivery of ink cartridges to users.



`Delivery/Collection Vehicle

A van or lorry delivers new ink cartridges, and old cartridges are collected from the ALL-INK box during the same trip, ensuring an efficient and coordinated logistics process.



ALL-INK SYSTEM

Quality Check



Retrieved ink cartridges are tested for quality to make sure they are fit for refilling and cap alterations if they haven't already happened.

Recycling Sorting



If the ink cartridge fails the quality check and is altered then its cap is removed. The cartridge is then sorted into the respective manufacturer and is packaged.

Recycling Vehicle



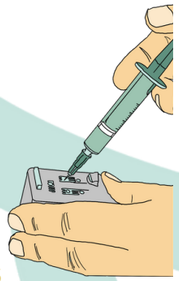
A van or lorry is employed to transport the sorted ink cartridges to their designated recycling centers, ensuring the proper disposal of empty ink cartridges.

Inventory



Once refilled the ink cartridges are organised in inventory sections by category, the ink cartridges are easily accessible when consumers need them. They are stored and ready for prompt delivery.

Refill Systems



Ink cartridges are carefully refilled by trained and skilled workers to ensure they are fit for high quality print.

ALL-INK Cap



If the quality check passes and the ink cartridge cap has not been altered, the caps are changed to ALL-INK caps to enhance the refill process, increasing overall efficiency.

Means of processing

Means of Processing (1) - In Production:
ALL-INK Cap is die cast using aluminium and the Box is injection moulded. For the ALL-INK Box the old ink cartridge cap is forcefully taken off and is repurposed to be used in the Box and foam from failed ink cartridges are cleaned and are then pressed down into the ALL-INK Box to form a layer of protection.

Means of Processing (2) - Pre-Consumption
Before use of the subscription the ALL-INK Box must be installed by consumers houses so they can use it to place their empty ink cartridges and so new ink cartridges can be placed into it.

Means of Processing (4) - Identifying and Detecting Pre- and Post-Consumption
The ALL-INK App identifies and detects the ink levels both before and during consumption utilising bluetooth technology. Users receive timely notifications through the app when ink levels are running low, which facilitates the need for transport.

Means of Processing (5) - Sorting For Recovery
When ink cartridges are brought back they are placed on conveyors and automated systems are used to categorise cartridges based on their original manufacturer. These automated systems also carry out quality checks.

Means of Processing (6) - Cleaning Post Consumption
When the quality checks pass there are strong magnets further along the conveyor that strip of the cap of ink cartridges. Old foam is then removed from the cartridge and new foam is inserted into the cartridge.

Means of Processing (7) - Refurbish To Recover
With new foam inserted then a part of the automated system will be used to refill the ink cartridge,.The cap will be placed back on the cartridge and will be stored in an inventory ready to be put back on the cycle.

Means of transportation

ALL-INK relies on a fleet of vans/lorries as its primary transportation mode. These vehicles efficiently collect empty ink cartridges and deliver new ones, transporting them to the ALL-INK manufacturing center. At this center, the cartridges undergo quality testing. If they pass, the vans/lorries initiate the process of delivering the cartridges back to users' houses. In the event of quality test failures, the vans/lorries redirect the cartridges to appropriate recycling centers. This meticulous transportation process ensures a closed-loop system for cartridge reuse and recycling within the ALL-INK ecosystem.

Means of communication

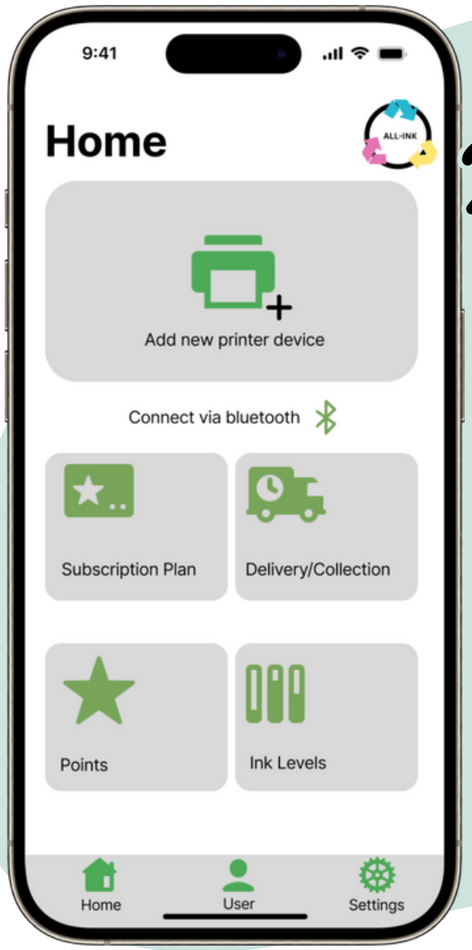
The ALL-INK system employs a robust communication setup. The ALL-INK app serves as the user interface, managing subscriptions and offering real-time delivery tracking. Bluetooth connectivity links the app with printers, ensuring continuous ink level monitoring. This streamlined communication framework enhances user experience, promotes efficient ink management, and supports the system's sustainability goals.

ALL-INK APP

Emmanuel Irechukwu

Detailing the centre point of all the infrastructure the ALL-INK application that facilitates the whole infrastructure system.

1



The ALL-INK app's homepage is a comprehensive guide, outlining the myriad capabilities it offers. This intentional design ensures users swiftly grasp the system's functionalities, setting clear expectations. From managing subscriptions to tracking deliveries, the homepage provides a quick understanding, empowering users with immediate insights into the full spectrum of possibilities within the ALL-INK system.

2



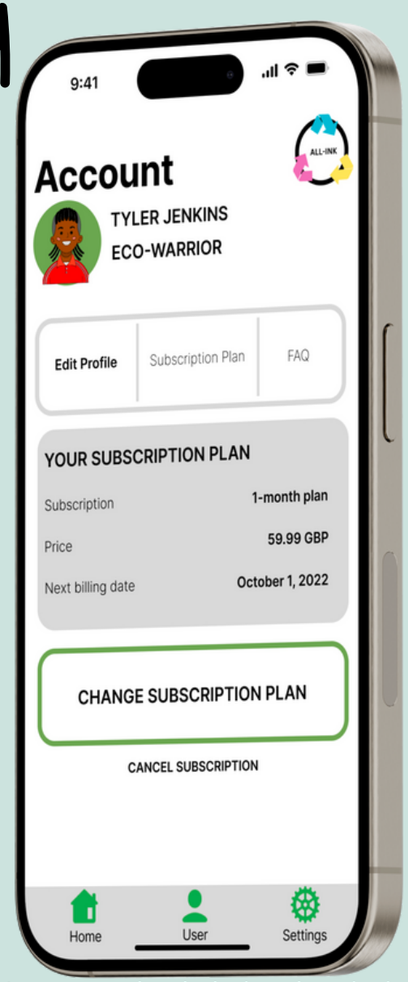
This is the delivery section of the app consumers will be able to book for a delivery. However deliveries will only be allowed to be scheduled if the ink cartridge is below a certain threshold. If the date is filled in this is date for coloured ink and if only the outline is coloured this is the day black ink cartridges are collected and delivered.

3



This shows users their ink cartridge levels this is useful because it allows for users to easily gauge when they need replacements. This will also help users to know the accurate subscription

4



Users can check their subscription plan to check their next billing date, also here they can check the amount they pay and they can decide to cancel their subscription or change their subscription plans.

5



This is the rewards page it has offers that users can redeem with their points to make their subscription plans cheaper, this incentivizes users to actually carry out the tasks that they are supposed to as they accumulate more points they unlock different badges and titles which make it game like for users.

Insight

The essence of the ALL-INK infrastructure transcends mere ink management; it meticulously engineers an interconnected ecosystem that not only addresses user needs but goes further to embrace sustainability at its core. This system is a testament to a thoughtful, holistic design approach, where every component is a deliberate piece in the puzzle of enhancing user experience, promoting eco-consciousness, and crafting a seamless, circular ink lifecycle.

Arancha Ramirez

PRODUCT ANALYSIS

ALL-INK system is compared to those available in the current market to analyse the product's sustainability. Additionally, we explored how consumers interact with our system through extensive interviews with printer owners.

ALL-INK

Description:
ALL-INK is a perfect combination of a high quality ink cartridge with low-carbon footprint and minimised plastic consumption and pollution. The subscription service makes it more convenient.

Service details:
- Return scheme keeps all of the waste in a closed looped system
- There is no packaging

The Recycling Factory Shop

Description:
The Recycling Factory buys and sells recycled ink cartridges. They provide a free recycling box online which the user must take to their local post office only once its full (50 cartridges). After recycled cartridges are processed, payment will take place [2]

Service Details:
- Only four brand's cartridges are accepted and not all cartridge models are eligible
- Cardboard packaging for every collection and delivery

CANON

Description:
Pixma Print subscription service offers multiple plans depending on the number of pages you print per month and automatically delivers ink when you start running low [3]

Service Details:
- Although Canon does offer a recycling scheme, it isn't included or advertised in the pixma print plan.
- Cardboard packaging for every delivery

Product and System Comparison

- User can customise delivery date of the cartridges via the app, which encourages recycling and a sustainable lifestyle
- The subscription offers a return scheme so that the consumer doesn't have to worry about disposing of it sustainably
- Scheme admits all ink cartridges
- Users are incentivised to recycle with additive points which they can exchange to lower their subscription prices
- Circular principles considered throughout the design

- Consumers sign-up through the website rather than an app
- The sign-up service offers a monetary return scheme
- Scheme admits limited consumables
- Users have monetary incentives to recycle (up to £2 per cartridge)
- Some circular principles present in the design

- Consumers subscribe through the website and manage their account online using *My Account*
- Canon has a free recycling scheme in place
- Scheme only accepts genuine Canon consumables
- Users don't have valuable incentives to recycle
- Some circular principles present in the design

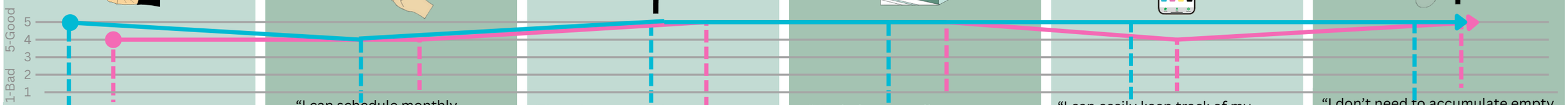
User Experience Map



Passionate Patrick (Age 32) eco-native
- Chose ALL-INK for its circularity
- Cares about and is proactive about sustainability
- Only uses printer twice per week



Convenient Claire (Age 26) eco-aware
- Chose ALL-INK for its practicality
- Cares about sustainability but isn't proactive
- Uses printer every day for work

	SUBSCRIBING	ORDERING	RECEIVING	USING	INTERACTING	RETURNING
Actions						
Experience (5-Good, 1-Bad)						
Thoughts	"I think the subscription will increase circularity and encourage the development of more sustainable products." "Its so much easier to keep my printer always stocked and working."	"I can schedule monthly deliveries so that I always have a constant supply of ink " "I can schedule everything wherever I am on my phone."	"There is no packaging I have to dispose of!" "I'm glad this service is delivery based. I no longer have to waste time travelling to the store."	"Works just as well as my previous cartridges. I'm very demanding with the quality of my documents." "Its very easy to insert in the printer. I don't have to do any work."	"I can easily keep track of my printing trends and needs." "I didn't have to check the printer for the ink levels. The app notified me, so I remembered to schedule a new delivery."	"I don't need to accumulate empty cartridges before sending them to be recycled." "I didn't have to order or purchase additional packaging. I don't have to travel anywhere, I just got a notification on when to deposit it outside my home!"
Design Intents	Consumer needs vary from user to user, so different subscription plans are available to suit everyone's needs.	Users can use the mobile app to order and schedule their deliveries no matter where they are. ALL-INK app provides easy communication of information between brand and consumers.	User's can't waste precious time, so providing tailored and flexible delivery options is essential. Additionally, ALL-INK boxes focus on durability and protection. Its sturdy metal surface protects the cartridges and the transparent polymer cover allows consumers to easily detect when a delivery has arrived	We carefully designed the ink cartridges so that ink quality remained unchanged and compatibility wasn't an issue, since user's second top priority and concern is the quality of the finished print.	ALL-INK app allows users to keep track of printing trends and remotely notifies them of their need to schedule a new delivery. Additionally, we encourage users to participate in our recycling scheme by providing incentives such as discount codes to earn a brand loyalty and promote a more sustainable lifestyle.	In response to customers needs, collection & delivery take place at the same time, to avoid clutter in the user's home and incorrect disposal of the ink cartridges.

Arancha Ramirez

PS SUSTAINABILITY & DATA SYSTEM

Further analysis using the Eco Audit tool is carried out on one cartridge case and lid. All lifecycle stages are considered and analysed and a material circularity diagram is derived. Additionally, consumer data and trends are explored.

Stored Data

- Thanks to our package-free system, 0.73kg of single-use plastic will be saved per user every year.
- Additionally, the CO2 footprint would decrease by 3.57kg per user every year.

Comparison

ALL-INK cartridge
Distance: 190km
Year carbon emissions: **2.91kg**

The Recycling Factory Shop
Distance: 213km
Carbon Emissions: **5.92kg**

CANON-Pixma Print
Distance: 1202km
Carbon Emissions: [4] **6.48kg**

Recycling is often considered the most sustainable option but reusing consumables is a more environmental friendly option as it reduces the demand for new raw materials, lowers energy consumption and carbon footprint and reduces waste generation . Additionally, re-filling ink cartridges it is more cost-effective than buying new ones.

Eco Audit

recycled Canon ink cartridge

refilled ALL-INK cartridge

Assumptions

- Average transport distance: 190km Birmingham to London
- The ink cartridge lid and case are replaced every year.
- The average subscription includes a monthly exchange, so the user goes through 12 ink cartridges.
- Energy required to clean the ink cartridge was not considered in the Eco Audit analysis

Energy Analysis

Material	**62.8%**	5.85MJ
	89.7%	4.11MJ
Manufacture	**11.1%**	1.05MJ
	9.9%	0.45MJ
Transport	**1.8%**	0.17MJ
	0.2%	0.01MJ
Use	**24.8%**	2.35MJ
	0%	0.0MJ
Disposal	**0.6%**	0.03MJ
	0.2%	0.01MJ
-2.2MJ		
-2.25MJ		

CO2 Footprint Analysis

Material	**52%**	0.231kg
	85.5%	0.207kg
Manufacture	**17.8%**	0.079kg
	14%	0.034kg
Transport	**2.7%**	0.012kg
	0.3%	0.001kg
Use	**26.7%**	0.119kg
	0%	0.0kg
Disposal	**0.8%**	0.004kg
	0.2%	0.001kg
-0.0789kg		
-0.0792kg		

End of Life Analysis

Recycled and refilled consumables were compared and a significant decrease in energy and CO2 footprint can be observed.

[4]

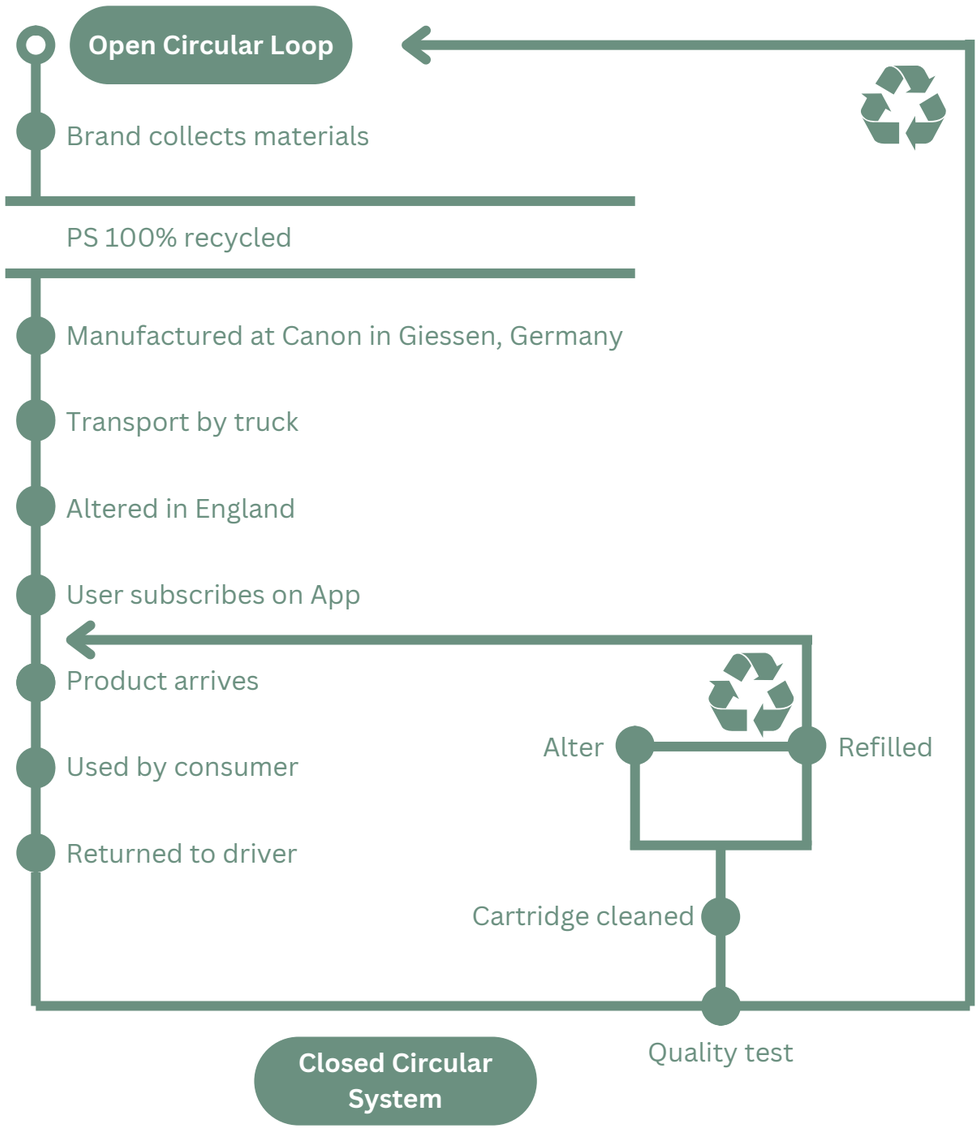
Results

ALL-INK has a smaller CO2 footprint, than competitors. Although their highest impact occurs during the material and manufacture processes they are still less than that of Canon's closed circular loop. Compared to other cartridge parts, the interlocking system has the highest CO2 footprint (1.6kg). This is due to the continued remanufacture and attaching of the components to the recycled cartridge.

The CO2 footprint of the disposal is very low since our product is **100% closed loop.**

The analysis indicates that 85.5% of CO2 emissions come form manufacturing, which is normal .compared to other ink cartridges in the market. ALL-INK's transport and disposal are **incredibly lower**, making our cartridge much **more sustainable** since manufacture can be avoided if the cartridge passes the quality check.

Material Circularity Analysis



VALUE SYSTEM

System Value Proposition

Closed loop recycling of ink cartridge

Bi-directional information flow

Resource flow

Forward finincial flow

LyondellBasell

Ecological benefits
- Recycled pp have a 80% lower carbon footprint than virgin equivalent.
- Recycling PP consumes 50% to 75% less energy than producing new PP from scratch.
- Recycling PP reduces the demand for new raw materials.
- Recycling PP diverts plastic waste from landfills and incineration, contributing to a reduction in the overall volume of plastic waste.
- By relying on recycled PP, which reduces the need for virgin resource extraction, there is less pressure on natural habitats, promoting biodiversity conservation.

Ecological costs
- Some recycling processes may involve the use of chemicals. These chemicals have potential effects on ecosystems and water quality.

Economic value proposition
Supply 100% recycled PP pellets
Help customer avoid plastic tax

Ecological value proposition
- Reduce the carbon footprint of supplied PP
- Preserves natural resources (petroleum oil)
- Help customers to achieve their sustainability goals
- Provide consumer high quality yet environmental friendly products

Social value proposition
- Reduce oil extraction site
- By reducing the need for new plastic production, minimizes the release of pollutants and toxins associated with the manufacturing process.
- Help customer comply some regulations in some regions.

Business pattern
selling TO

Customer relationships
- Supply Chain Integration:
  - Work towards integrating supply chain with manufacturers. Understanding their production schedules and adapting supply accordingly.
- After-Sales Support:
  - Address any issues promptly and offer assistance with any technical or logistical challenges.
- Transparency:
  - Be transparent about sourcing and recycling processes.
- Quality Assurance:
  - Provide high-quality recycled PP that meets or exceeds industry standards.

Information
Quantity and quality of scrap PP

Business activities
- Collection and Sorting scrap PP
- Cleaning scrap PP
- Shredding scrap PP
- Melting Shredded PP
- Extruding PP pellets
- Quality Control test

Information
material properties needed for PP

Internally-used resources
- Human Resources:
  - Recycling Specialists
  - Research and Development Team
  - Workers
- Technological Resources:
  - Recycling Equipment
- Financial Resources:
  - Capital for Equipment Investment
  - Operational Budget
- Intellectual Property:
  - Recycling Process Patents
- Physical Resources:
  - Recycling Facility
  - Storage Facilities
- Water Resources
- Energy Resources

Core resource
Ownership
Recycled PP pellets

Cost

To	Description
	LyondellBasell buys disassembled ink cartridge as scrap PP from All-ink

Revenue

From	Description
	LyondellBasell receives payment from Cannon for PP pellets

Partners
- Dow Inc
- NOVA Chemicals Corporation

Social benefits
- Health and Well-being: Reduced pollution from plastic waste minimizes the risk of contamination in soil, water sources, and food chains, consequently benefiting human health and well-being.
- Job Creation and Economic Opportunities: The recycling industry generates employment opportunities. Collection, sorting, processing, and manufacturing recycled PP products require a workforce, potentially creating jobs in various sectors.

Social costs
- Inequitable Distribution of Benefits: The benefits of recycling may not be equally distributed across communities. In some cases, certain communities may bear the burden of waste disposal and recycling facilities, leading to environmental justice concerns.

Canon

Ecological benefits
- By incorporating recycled PP into ink cartridges, Canon can contribute to lower overall emissions.
- Divert plastic waste from landfills and incineration.
- Using recycled PP can position Canon in compliance with these evolving regulations.

Ecological costs
- Transporting recycled materials to manufacturing facilities can contribute to greenhouse gas emissions.

Economic value proposition
Supply customers with ink cartridges capable of generating high-quality prints.

Ecological value proposition
- Reduce the volume of plastic in landfills and oceans
- A lower carbon footprint compared to cartridges made from virgin plastics.
- Eco-conscious printing without compromising quality.
- Help consumers to live a more sustainable lifestyle and achieve personal sustainability goal.

Social value proposition
- Participates in closing the loop of plastic use, encouraging the recycling and reuse of materials in a continuous cycle.
- Demonstrates Cannon's commitment to sustainability and environmental stewardship. This can enhance the brand image and reputation
- Encourage suppliers to adopt environmentally friendly practices, creating a ripple effect of positive environmental impact.

Business pattern
selling TO

Customer relationships
- Reliable Supply Chain:
  - Ensure a steady and reliable supply of ink cartridges.
- Customized Solutions:
  - Offer a range of ink cartridge products that cater to the specific needs of the market.
- Training and Technical Support:
  - Provide training and technical support to customers.
- Marketing Support:
  - Collaborate on marketing initiatives to promote ink cartridges.Support for advertising efforts.
- Inventory Management:
  - Forecasting demand, managing stock levels, and coordinating logistics to minimize stock outs or overstock situations.
- Responsive Customer Service:
  - Provide responsive customer service to address any issues or concerns promptly.
- Long-Term Partnership Development:
  - Foster a long-term partnership mindset. Consider the distributor as a strategic partner, and work collaboratively to achieve mutual business objectives.

Information
Required quantity of PP pellets

Business activities
- Research and Development to innovate and improve the design, functionality, and efficiency of ink cartridges.
- Set up and manage manufacturing facilities to produce ink cartridges in large quantities.
- Molding the plastic body of the ink cartridge.
- Assemble necessary nozzles and chips on plastic body.
- Filling ink.
- Quality Control.
- Distribute to retailers, wholesalers, and end-users.
- Provide customer support for inquiries, technical issues, and warranty claims.

Information
Ink cartridge pricing and functionalities

Internally-used resources
- Human Resources:
  - Design and Engineering Teams
  - Research and Development Teams
  - Production and Manufacturing Staff
  - Quality Control and Testing Teams
- Technological Resources:
  - Manufacturing Equipment
  - Testing Equipment
- Financial Resources:
  - Capital for Research and Development
  - Operational Budget
- Intellectual Property:
  - Patents related to ink cartridge production
- Physical Resources:
  - Manufacturing Facilities
  - Warehousing
  - Distribution Centers
- Water Resources
- Energy Resources

Core resource
Ownership
packed ink cartridges

Cost

To	Description
	Cannon buys PP pellets from LyondellBasell

Revenue

From	Description
	Cannon receive payment from All-ink for ink cartridges
	Cannon receive payment from user for their ink cartridge order

Partners
- Dell technologies
- ThompsonKnight
- Shell plc

Social benefits
- The process of collecting, sorting, and processing recycled PP to manufacture ink cartridges can create jobs in various stages of the supply chain.
- The use of recycled materials in ink cartridge manufacturing provides an opportunity for educational initiatives.This can raise awareness about the importance of recycling, waste reduction, and sustainable practices.
- The demand for recycled materials can create opportunities for social enterprises involved in waste collection and recycling.
- The use of recycled materials in manufacturing can generate less pollution.Contributing to better air and water quality, positively impacting the health and well-being of communities surrounding manufacturing facilities.

Social costs
- Waste Export Issues: There may be a tendency to export plastic waste to developing countries for processing. This can lead to environmental and health hazards in those regions, creating social justice concerns and negative impacts on communities.
- Stress on small business: Small enterprises often possess limited technological capabilities and resources for processing recycled materials. Should the market perceive recycled materials as superior, this may potentially marginalize small businesses, leaving only larger entities with significant market influence.

ALL-ink

Ecological benefits
- Reduced Waste: Customers receive ink cartridges based on their usage, reducing the chance of overbuying and wasting.
- Reduced Packaging: reducing the need for excessive packaging often associated with individual purchases.
- Minimizing the need for transportation cartridges. Fewer transportation activities contribute to lower greenhouse gas emissions.
- Economic Sustainability:Subscription models promote longer-term relationships with customers.Increasing brand loyalty, allowing the company to plan for the long term and invest in sustainable practices.

Ecological costs
- Some components of ink cartridges are not biodegradable. There may still be challenges in creating fully biodegradable products.
- Implementing recycling programs requires resources for collection, transportation, and processing of returned cartridges.

Economic value proposition
- Provide customers with subscription services offering unlimited access to ink cartridges capable of producing high-quality prints.
- Monitor customers' ink levels, ensuring they no longer need to concern themselves with the possibility of running out of ink.

Ecological value proposition
- Significantly less packaging
- Empower customers to make environmentally conscious choices by providing affordable yet sustainable services.
- Minimize carbon emissions associated with the business operations.

Social value proposition
- Creation of employment opportunities and commitment to fair labor practices.
- Actively engage with and contribute to local communities.
- Prioritize the health and safety of employees and customers. Adhere to high standards in workplace safety.Increase employee health and well being.
- creating opportunities for entrepreneurship, skills training in areas that may benefit from additional support.

Business pattern
provide service To

Customer relationships
- Communication and Transparency:
  - Maintain open and transparent communication with customers. Clearly communicate how the subscription service works, including details about billing, delivery schedules, and any changes in the service.
- Personalization:
  - Tailor subscription plans to match individual requirements, ensuring that customers receive the right amount of ink based on their usage.
- Customer Support:
  - address queries, concerns, and technical issues promptly.
- Problem Resolution and Compensation:
  - Handle customer complaints and issues effectively. Provide compensation or solutions when problems arise.
- Community Engagement:
  - Create forums, social media groups, or events where customers can interact, share tips, and discuss their experiences.
- Security and Privacy:
  - Assure customers that their data is handled securely, and their privacy is respected.

Information
Quantities of ink cartridge supplied

Business activities
- Ink Cartridge Subscription Service: Offer a subscription model where customers receive regular shipments of ink cartridges based on their usage.
- Store both used and new ink cartridges
- Delivery of new or refilled ink cartridges
- Modify ink cartridge for easy refilling
- Quality Control.
- Eco-Friendly Product Design: Invest in research and development to create ink cartridges with Eco-friendly materials.
- Online Platform and Subscription Management: Develop an easy-to-use online platform for customers to manage their subscriptions, track usage, and make adjustments to their plans.
- Community Engagement Programs: Engage in community programs that support education, environmental conservation, or other social causes.
- Partnerships with Environmental Organizations: Form partnerships with reputable environmental organizations to amplify the company's commitment to sustainability. Collaborate on initiatives that contribute to environmental conservation.
- Customer Loyalty Programs: Implement customer loyalty programs that reward long-term subscribers with discounts, exclusive offers, or other perks.
- Continuous Innovation: Stay abreast of technological advancements in ink cartridge technology and sustainable materials. Strive for continuous innovation to improve the environmental performance of products and services.

Information
- Demand of ink cartridge from users
- User's ink cartridge using pattern
- Delivery address from the users

Internally-used resources
- Human Resources:
  - Design and Engineering Teams
  - Quality Control and Testing Teams
  - Delivery/Collection teams
  - Customer Service Teams
- Technological Resources:
  - Manufacturing Equipment for modifying ink cartridge
  - Testing Equipment
- Financial Resources:
  - Capital for initial set ups
  - Operational Budget
- Physical Resources:
  - Manufacturing Facilities
  - Warehousing
  - Distribution Centers
- Information Technology (IT):
  - Software Systems for inventory management, order processing, and quality control.
  - APP for user to manage subscriptions
- Water Resources
- Energy Resources

Core resource
Ownership
modified ink cartridges

Cost

To	Description
	All-ink buys new ink cartridge from Cannon
	All-ink covers the initial modification costs for the ink cartridge.

Revenue

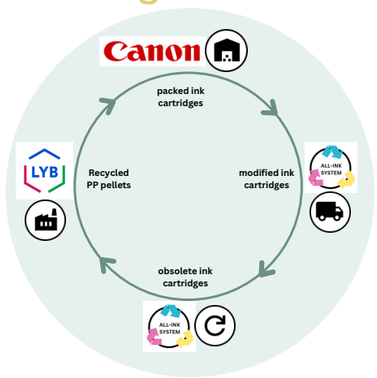
From	Description
	users pay for the unlimited ink cartridge service to All-ink

Partners
- RECOUP
- Cannon
- SEA life trust

Social benefits
- By offering affordable ink cartridges, make printing solutions more accessible to students, small businesses, and non-profit organizations.
- Promoting diversity and inclusion within the company can set an example for other businesses. Fostering an inclusive workplace.
- The subscription model and excellent customer service contribute to customer satisfaction and convenience. This, in turn, positively impacts the daily lives of customers who rely on printing solutions for various purposes.
- By conducting the majority of delivery and collection activities during off-peak hours,Alleviating traffic stress and save commuting time for individuals utilizing the roadways.

Social costs
- Reliance on traditional printers and ink cartridges may contribute to the digital divide. Some communities or individuals may be left behind as technology evolves, affecting their access to information and services.
- Automation and decrease in need for new inc cartridges may lead to job displacement, impacting individuals who were previously employed in traditional printing-related industries.

System resource flow diagram :



system drivers

- Regulatory Compliance: Governments and regulatory bodies may impose stricter regulations on the use of virgin plastics and promote the use of recycled materials.
- Consumer Demand and Perception: Growing consumer awareness and concern about environmental issues, particularly plastic pollution, can drive demand for sustainable products.
- Cost Efficiency: Recycled PP can potentially be a more cost-effective option, especially as recycling technologies continue to improve.
- Supply Chain Resilience: Dependence on finite fossil fuels for virgin plastics may pose risks.Recycled materials provide an alternative source.
- Long-Term Resource Security: Anticipating potential future scarcity of virgin resources, invest in recycling processes to ensure a long-term, secure supply of raw materials for production.

system drivers

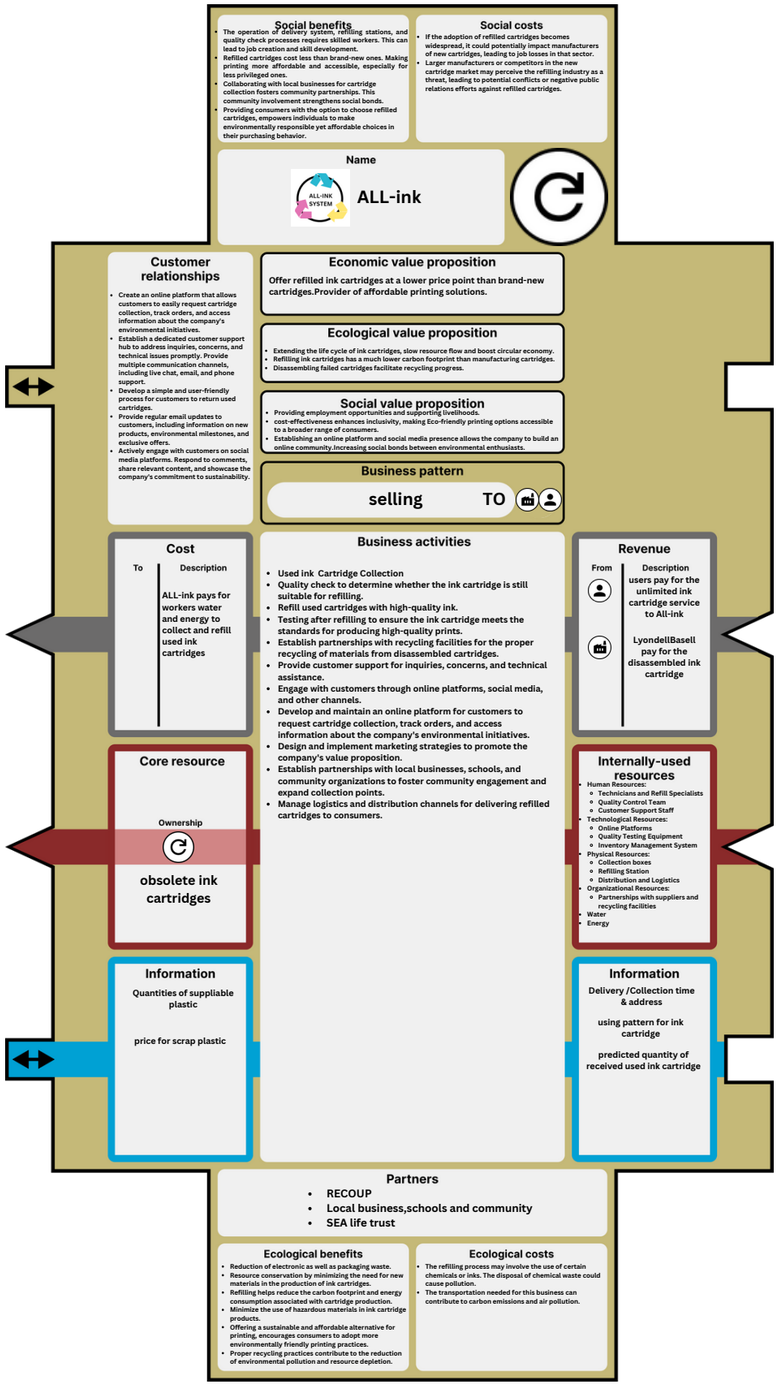
- Regulatory Future Compliance: Governments and regulatory bodies may impose stricter regulations on the use of virgin plastics and promote the use of recycled materials.
- Consumer Perception and Brand Image:Many consumers today are environmentally conscious and prefer products with a lower environmental impact
- Supply Chain Resilience: Depending solely on virgin plastic may pose risks due to fluctuations in raw material prices or supply chain disruptions. Using recycled PP can provide a more diversified and resilient supply chain.
- Innovation and Competitive Advantage:Adopting recycled materials can set a company apart from competitors, creating a unique selling proposition in the market. Innovation in materials can lead to new product development and a competitive edge.

system drivers

- Steady Revenue Stream: Subscription models provide a predictable and recurring revenue stream.
- Customer Retention:Companies can increase customer retention rates and lifetime customer value, reducing the need to constantly acquire new customers.
- Enhanced Customer Experience: Subscriptions can be designed to provide a more seamless and personalized customer experience.
- Data and Insights: Subscription models generate valuable customer data and insights. Companies can leverage this data to better understand customer behavior, preferences, and usage patterns, enabling them to make informed business decisions and improve their offerings.
- Consumer Behavior shift: With the rise of the subscription economy, consumer preferences have shifted towards services and experiences over ownership.

VALUE SYSTEM

Chuwei Xiong



system drivers

- Refilling can be a unique selling point, setting the company apart from competitors and attracting environmentally conscious consumers.
- Monitoring market trends and industry shifts towards sustainability can be a driving factor. The market is moving towards Eco-friendly alternatives.
- Economic benefits of refilling compared to recycling. Refilled ink cartridge having much higher value compare to simply recycling.
- Advocate for a strengthened corporate social responsibility (CSR) strategy.
- Potential positive impact on employee morale and engagement. Participating in environmentally responsible initiatives like refilling can boost employee pride and satisfaction, contributing to a positive work culture.
- A sustainable and Eco-friendly initiative like refilling can resonate positively with consumers, leading to long-term customer relationships.
- With anticipated stricter regulations and heavier taxes on plastic products, refilling has the potential to align with future regulatory requirements, offering both compliance and tax savings.

	Benefit	Costs	Cost mitigation
Finances	• LyondellBasell receives payment from Cannon for PP pellets • Cannon receive payment from All-ink for ink cartridges • Cannon receive payment from user for their ink cartridge order • users pay for the unlimited ink cartridge service to All-ink • LyondellBasell pay for the disassembled ink cartridge	• LyondellBasell buys disassembled ink cartridge as scrap PP from All-ink • Cannon buys PP pellets from LyondellBasell • All-ink buys new ink cartridge from Cannon • All-ink pays for delivery/collection of ink cartridges to customers • All-ink covers the initial modification costs for the ink cartridge • All-ink pays for workers water and energy to collect and refill used ink cartridges	• Cannon conduct a price comparison, identifying the most cost-effective and usable plastic available. • All-ink has designed the simplest modification process to minimize the cost associated with enhancing ink cartridges for refilling. • All-ink reducing disassembly costs through a purposefully designed, disassemble-friendly structure for ink cartridges. • All-ink secures the ink cartridge for extended refilling cycles, mitigating the need for frequent purchases of new cartridges.
Ecology	• Refilling ink cartridges, as opposed to manufacturing new ones, can result in energy savings exceeding 91%, thereby making a substantial contribution to reducing carbon emissions. • By refilling contributes to the conservation of raw materials and resources that would be used in the production of new cartridges. This helps decrease the demand for new plastics and metals. • The extraction of raw materials for manufacturing cartridges can contribute to habitat destruction and biodiversity loss.Minimizing the demand for new cartridges can contribute to the preservation of ecosystems. • Refilling ink cartridges reduces the amount of plastic waste generated from disposable cartridges. • Refilling slows the resource flow, fostering a circular economy.	• The transportation required for collecting/delivering ink cartridges to customers has the potential to generate additional carbon emissions and contribute to air pollution. • Ink, when not handled carefully during the refilling process, poses the risk of causing environmental pollution.	• Utilizing online platforms for coordination minimizes failed delivery attempts. Implementing a collection and delivery in a single trip further reduces transportation needs, enhancing efficiency. • Careful management of ink and the responsible disposal of used ink to minimize chemical pollution. • Continuously investing in innovation for biodegradable ink materials, fostering research for new and more sustainable ink solutions.
Society	• The process of collecting, sorting, and processing ink cartridges can create jobs in various stages of the supply chain. This can contribute to local economic development and provide employment opportunities. • Recycling initiatives involving local communities in the collection and recycling process, can empower individuals and communities. Community members may feel a sense of ownership and responsibility for environmental stewardship, leading to increased engagement. • The online platform can serve as a virtual hub for environmental enthusiasts, fostering social bonds among them and inspiring more individuals to become environmentally conscious. • The app and the collection process present an excellent opportunity for educational initiatives to inform and raise awareness about environmental concerns among people. • By providing more sustainable products at a lower price compared to new ink cartridges,break the stereotype that environmentally friendly items must be more expensive, challenging preconceptions and encouraging widespread adoption of sustainable practices. • Lowering the price for printing contributes to addressing rising social inequality by providing individuals with fewer privileges access to more affordable printing options.	• In targeting ink cartridge sales, major manufacturers of ink cartridges may perceive ALL-ink as a potential rival, potentially leading to conflicts. This competitive dynamic might prompt significant manufacturers to employ strategies such as technical obsolescence. • The shift towards subscription-based models may be more challenging for individuals or communities with limited access to digital technologies. It could contribute to a digital divide, where those without adequate technology or internet access are excluded. • The benefits of recycling may not be equally distributed across communities. In some cases, certain communities may bear the burden of waste disposal and recycling facilities, leading to environmental justice concerns. • The adoption of automated or more efficient recycling/refilling technologies may reduce the need for manual labor in certain stages of the process, potentially leading to unemployment for some workers.	• Form alliances and partnerships with printer manufacturers or third-party technology providers. Collaborative agreements can lead to mutual benefits, ensuring that ALL-ink remains well-integrated with evolving printer technologies. • Implement digital literacy programs and offer resources to help individuals and businesses adapt to digital advancements. • Work towards ensuring that recycling facilities and used ink cartridge collection are distributed equitably across communities. Collaborate with local authorities and organizations to establish recycling centers in under-served areas, addressing disparities in access to recycling services. • Allocate resources to invest in workforce training programs. Provide training opportunities for employees whose roles may be impacted by technological advancements.

Challenge faced by the existing refilling business	All-ink's solution
• Inconsistent quality may lead to poor printouts or even damage to printers	• Conducting a comprehensive quality check both before and after the refilling process to ensure that all cartridges meet the requisite standards for high-quality printing.
• Managing a steady supply of high-quality ink and compatible cartridges is crucial. Supply chain disruptions, due to raw material shortages or transportation issues, can affect business .	• Implementing a subscription-based business model allows ALL-ink to retain ownership of all ink cartridges. This approach facilitates the collection of used cartridges from users, ensuring a consistent and reliable supply chain.
• The market for ink cartridges is competitive. Standing out in a crowded market requires effective marketing and differentiation	• ALL-ink offers products that are both economically viable and environmentally sustainable, distinguishing itself from original manufacturers and establishing a unique value proposition in the market. • Leveraging subscription services and delivering excellent service quality to enhance customer loyalty.

PRODUCT & PACKAGING DESIGN

The product and packaging are shown below the materials and manufacturing methods used are documented, and the design for delivery box has also been stated.

ALL-INK Box

The ALL-INK Box, stationed outside users' homes, streamlines ink cartridge management. Users deposit their empty cartridges when depleted and receive new ones in the same box during deliveries. This approach eliminates excess packaging and simplifies the recycling process, requiring users only to place their empty cartridges into the ALL-INK Box.

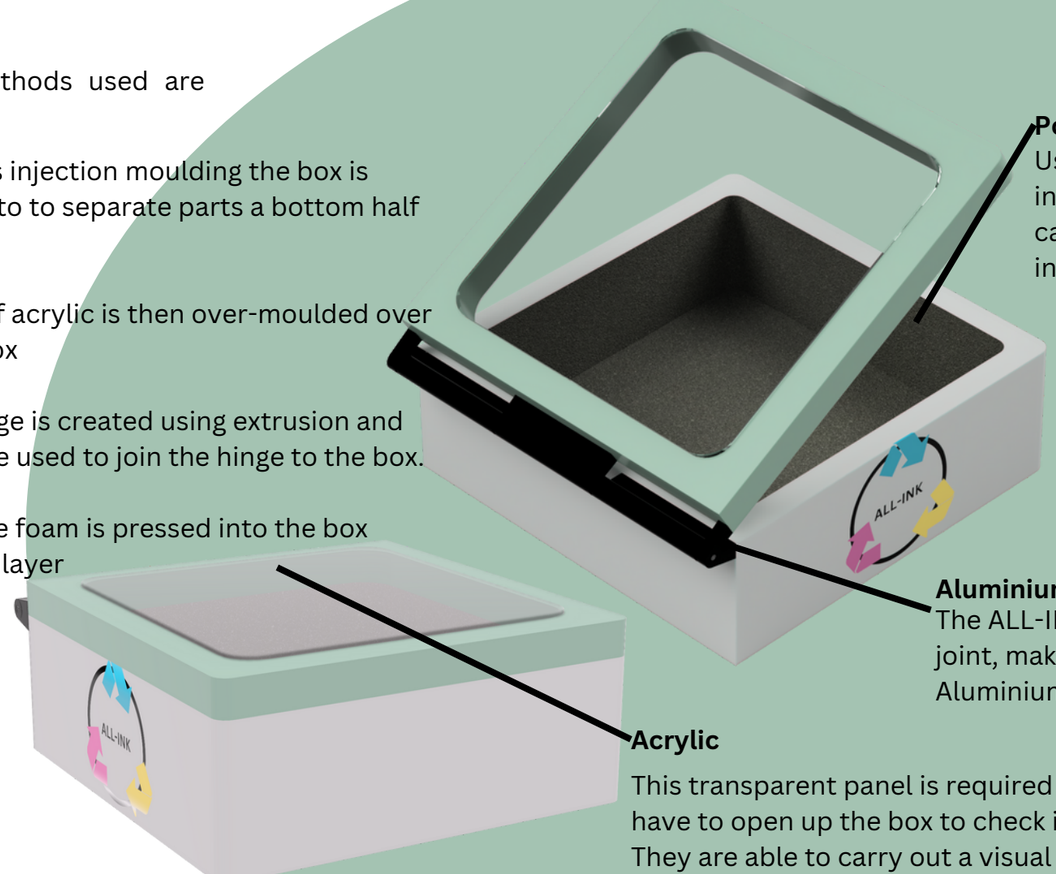
Manufacturing

1) The first process is injection moulding the box is injection moulded into to separate parts a bottom half and a top half.

2) A very thin layer of acrylic is then over-moulded over the top half of the box

3)The Aluminium hinge is created using extrusion and self taping screws are used to join the hinge to the box.

4) Used polyutherane foam is pressed into the box forming a protective layer



Polyutherane

Used Ink cartridge foam is placed into the box to protect the ink cartridge chips, also to provide insulation.

Aluminium

The ALL-INK Box is designed with a hinge joint, making it easy and smooth to open. Aluminium is a sustainable

Acrylic

This transparent panel is required as it means that users don't have to open up the box to check if a delivery has been made. They are able to carry out a visual inspection from a birds eye view.

Design For Reuse

The ALL-INK Box is crafted for reuse, promoting sustainability. Its durable design and materials support multiple use cycles, allowing users to consistently deposit and retrieve ink cartridges. This emphasis on reuse aligns with eco-friendly practices, making the ALL-INK Box an environmentally conscious choice for users focused on waste reduction.

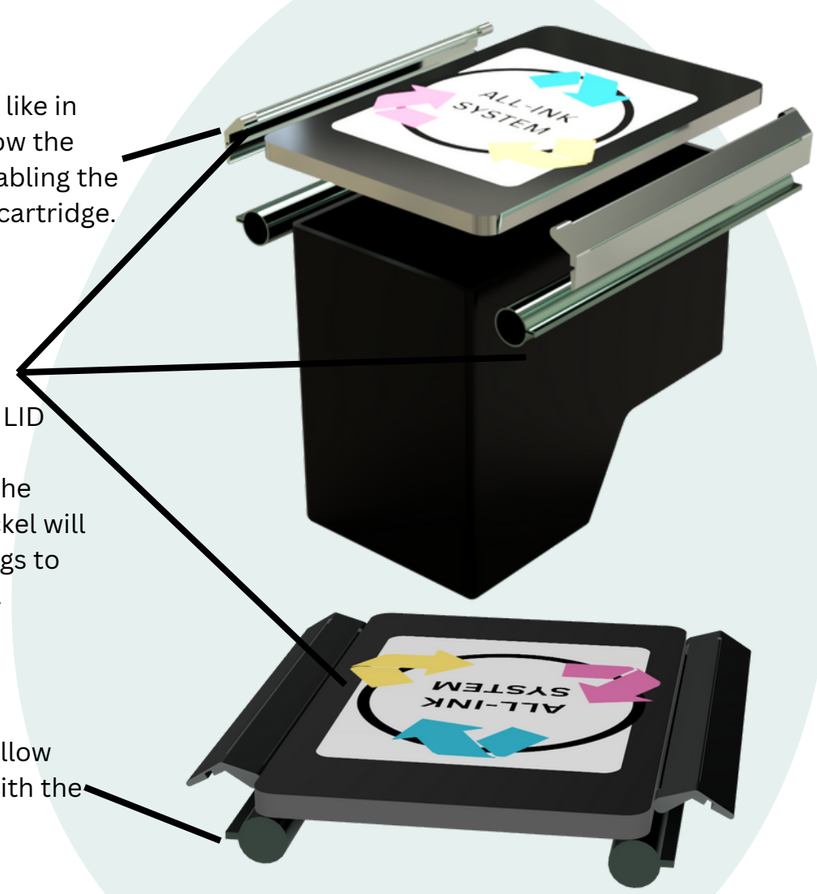
ALL-INK Cap

A hinge joint is utilised like in the ALL-INK Box to allow the legs to move freely enabling the interlocking of the ink cartridge.

Aluminium

The component of the LID will be made out of aluminium but where the interlocking occurs nickel will be used to allow the legs to magnetically interlock.

Magnets are used to allow the legs to interlock with the ink cartrisge body.



Manufacturing

1) The cap is die cast, and the interlocking parts are coated with nickel in order to allow for magnetism.

The ALL-INK cap features an innovative interlocking system for efficient ink cartridge refilling. The cap, equipped with legs, securely interlocks with the cartridge utilising magnetism, ensuring a tight seal to prevent leaks. This design promotes time and cost savings, facilitates cartridge reuse, and contributes to environmental friendliness by reducing reliance on single-use plastics.

Design For Disassembly

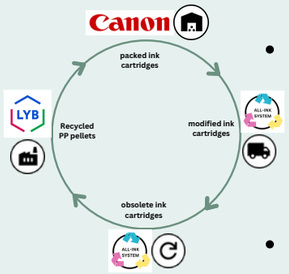
The ALL-INK cap is purposefully designed for easy disassembly, primarily to facilitate the professional refilling process managed by ALL-INK. Its components can be effortlessly separated, allowing trained technicians to access and refill the ink cartridges with precision. This thoughtful design streamlines the maintenance and servicing performed by ALL-INK, ensuring a seamless and reliable experience for users who subscribe to the ink cartridge service.

Design For Recycling

The ALL-INK cap is intentionally designed for seamless recycling by prioritising key elements. Crafted from easily recyclable aluminium, the cap adopts a single-material design, simplifying the recycling process. Engineered with a user-friendly snap-fit or screw mechanism, disassembly is effortless, promoting efficient recycling.

OUR SOLUTION

Value & Reusability



- **Interlocking mechanism ensures lid is tightly closed to avoid ink waste.**
- Offering a free-flowing ink service rather than physical ink cartridges, while retaining ownership of the cartridges under ALL-ink, ensures a heightened retrieval rate for used ink cartridges.
- Opting for refilling rather than recycling of usable ink cartridges maximizes the recovery of value from a used ink cartridge.

Product's Disassembly



- **To simplify access to the ink compartments, the cartridge lid is designed to be detachable, separating the body into three parts at manufacturer-level**
- The first is the lid. It seals the cartridge and contains the cartridge's code and ink information.
- The inside chamber contains the ink
- The case protects the ink inside and is joined to the lid in a latch interlocking mechanism

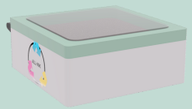
Collection & Delivery Subscription



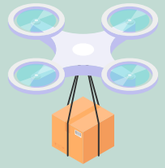
- **ALL-INK's collection and delivery system lets the consumer purely focus on the use and disposal of the cartridge**
- After the user empties the cartridge, they order a new one
- The user simply returns the empty cartridge by depositing it in the provided box for the disposal team to collect when they deliver the new one.
- Two additional incentives are given to encourage subscribers to remember to return their cartridges.
- First they will be given points which they accumulate and exchange for cheaper subscription plans.
- Additionally, the app is structured like a game where they unlock different badges and titles for user to become more engaged.

Feasibility & Time Horizon









0 — 5 — 10 years

- ALL-INK develops design, manufacturing process and system for product
- Collection & delivery system are explored

- Consumables are made in adequate dimensions and interlocking mechanism works smoothly
- Online platform and package-free product established, increasing brand loyalty
- Agreements with brands are established

- ALL-INK is implemented in multiple countries
- The app works flawlessly
- ALL-INK plans to reduce their carbon emissions by implementing drone delivery services and explores alternative bio-material inks

Summary & Novelty Highlights

ALL-INK service company pilots with its innovative subscription model in 5-10 years time. To design this cartridge and system with **maximised sustainability**, ALL-INK analysed key problems at the **consumer**, **product** and **system levels**. This encompasses the necessity of **high quality ink**, the opportunities to **combine** and **streamline delivery** and **collection services**, the **simplification** of cartridge **disassembly** and **incentives** to promote recycling.
ALL-INK cartridges have three main components: the **detachable lid**, the **ink chamber** and the **case**. We **prioritize refilling** cartridges as long as they uphold quality standards, as opposed to immediate recycling, making it part of a **mostly closed loop circular system**. This approach allows us to extract considerably more value compared to traditional single-use methods and significantly decrease carbon emission due to **transport** and **energy consumption** during **manufacturing.**
After comparing ALL-INK with similar services like Canon and The Recycling Factory, it remains the most sustainable after 1 year and the **only** service offering **delivery and collection** in the printing industry.

References

[1] therecylcingfactory.com-Cartridge Facts, Reasons why whe recycle cartridges [online] Available at: https://www.therecyclingfactory.com/facts
[2] therecyclingfactory.com- We buy and recycle ink cartridges [online] Available at: https://www.therecyclingfactory.com/
[3] www.canon.co.uk- Canon printer ink subscription, Pixma Print Plan [online] Available at: https://www.canon.co.uk/printer-ink-subscription/pixma-print-plan/
[4] Arancha Ramirez, Submission 1.1